

SGA 4, Exposé IV. Topoi

Draft translation, Section 10

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Translator's status note. This is a standalone fresh-only repackaging of the missing Batch 015 material. The translated range corresponds to source lines 8246–8573 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current mathematical English: *geometric morphism* for *morphisme de topos*, *internal Hom* for *faisceau de morphismes* where appropriate, *sheaf of morphisms* for the named object $\mathcal{H}om_E(X, Y)$, and *localization morphism*. Numbering follows the source.

10 Sheaves of morphisms

Proposition 10.1. Let E be a topos, and let X and Y be two objects of E . The functor

$$Z \mapsto \mathrm{Hom}_E(Z \times X, Y) \simeq \mathrm{Hom}_{E/Z}(X_Z, Y_Z)$$

is representable.

Indeed, inductive limits are universal in E (II, 4.3). Hence the functor $Z \mapsto Z \times X$ commutes with inductive limits. It follows that $Z \mapsto \mathrm{Hom}_E(Z \times X, Y)$ carries inductive limits in the argument Z into projective limits. It is therefore representable (IV, 1.4 and 1.2, equivalently 1.2.1).

10.2

The object representing the functor $Z \mapsto \mathrm{Hom}_E(Z \times X, Y)$ is denoted

$$\mathcal{H}om_E(X, Y),$$

or simply $\mathcal{H}om(X, Y)$, and is called the *sheaf of morphisms from X to Y* . It is bifunctorial in X and Y . Thus one has a trifunctorial isomorphism

$$\mathrm{Hom}_E(Z, \mathcal{H}om_E(X, Y)) \simeq \mathrm{Hom}_E(Z \times X, Y). \quad (10.2.1)$$

It follows from (10.2.1) that the bifunctor $(X, Y) \mapsto \mathcal{H}om(X, Y)$ sends inductive limits in the argument X (respectively projective limits in the argument Y) to projective limits in E .

Proposition 10.3. Let $v : E \rightarrow E'$ be a geometric morphism. For a variable object X of E and a variable object Y of E' , there is a bifunctorial isomorphism

$$v_* \mathcal{H}om_E(v^* Y, X) \simeq \mathcal{H}om_{E'}(Y, v_* X). \quad (10.3.1)$$

Proof. For every object Z of E' there is a chain of isomorphisms, functorial in all arguments:

$$\begin{aligned} \mathrm{Hom}_{E'}(Z, v_* \mathcal{H}om_E(v^* Y, X)) &\simeq \mathrm{Hom}_E(v^* Z, \mathcal{H}om_E(v^* Y, X)) && \text{(adjunction)} \\ &\simeq \mathrm{Hom}_E(v^* Z \times v^* Y, X) && \text{by (10.2.1)} \\ &\simeq \mathrm{Hom}_E(v^*(Z \times Y), X) && \text{since } v^* \text{ is left exact} \\ &\simeq \mathrm{Hom}_{E'}(Z \times Y, v_* X) && \text{(adjunction)} \\ &\simeq \mathrm{Hom}_{E'}(Z, \mathcal{H}om_{E'}(Y, v_* X)) && \text{by (10.2.1).} \end{aligned}$$

Thus the two sides of (10.3.1) represent isomorphic functors.

Corollary 10.4. Let C be a $\mathcal{U}$ -site, X a $\mathcal{U}$ -presheaf on C , Y a $\mathcal{U}$ -sheaf on C , and $\mathcal{V}$ a universe containing $\mathcal{U}$ such that C is $\mathcal{V}$ -small. Let $\widehat{C}_{\mathcal{V}}$ be the topos of $\mathcal{V}$ -presheaves on C , and let $C_{\mathcal{U}}^{\sim}$ be the topos of $\mathcal{U}$ -sheaves on C . Then the presheaf

$$\mathcal{H}om_{\widehat{C}_{\mathcal{V}}}(X, Y)$$

is a $\mathcal{U}$ -sheaf. If $X \rightarrow \underline{a}X$ is the canonical morphism from X to its associated sheaf, the corresponding morphism

$$\mathcal{H}om_{\widehat{C}_{\mathcal{V}}}(\underline{a}X, Y) \longrightarrow \mathcal{H}om_{\widehat{C}_{\mathcal{V}}}(X, Y)$$

is an isomorphism. Moreover there is a canonical isomorphism

$$\mathcal{H}om_{\widehat{C}_{\mathcal{V}}}(\underline{a}X, Y) \simeq \mathcal{H}om_{C_{\mathcal{U}}^{\sim}}(\underline{a}X, Y).$$

10.4.1

First suppose that C is a small site and that $\mathcal{V} = \mathcal{U}$. There is then a geometric morphism

$$C_{\mathcal{U}}^{\sim} \longrightarrow \widehat{C}_{\mathcal{U}}$$

(IV, 4.8), and (10.3.1) gives the assertions in this case. To pass from this to the general case, first observe that the associated-sheaf functor is independent of the universe (II, 3.6), and that the topos of $\mathcal{V}$ -sheaves on C is equivalent to the topos of $\mathcal{V}$ -sheaves on $C_{\mathcal{U}}^{\sim}$ (III, 4 and IV, 1). It remains only to prove the following lemma.

Lemma 10.4.2. Let E be a $\mathcal{U}$ -topos, let $\mathcal{V}$ be a universe containing $\mathcal{U}$, let $E_{\mathcal{V}}^{\sim}$ be the topos of $\mathcal{V}$ -sheaves on E , and let X, Y be two objects of E . There is a canonical isomorphism

$$\mathcal{H}om_E(X, Y) \simeq \mathcal{H}om_{E_{\mathcal{V}}^{\sim}}(X, Y).$$

10.4.3

The functor $\epsilon_E : E \rightarrow E_{\mathcal{V}}^{\sim}$ is fully faithful and left exact. Therefore, for every object Z of E , there is an isomorphism

$$\mathrm{Hom}_{E_{\mathcal{V}}^{\sim}}(\epsilon_E Z, \epsilon_E \mathcal{H}om_E(X, Y)) \simeq \mathrm{Hom}_{E_{\mathcal{V}}^{\sim}}(\epsilon_E Z \times \epsilon_E X, \epsilon_E Y).$$

But every object of $E_{\mathcal{V}}^{\sim}$ is an inductive limit of objects coming from E ; by (10.2.1) this proves the assertion.

10.5

Let $v : E \rightarrow E'$ be a geometric morphism. We shall define four bifunctorial morphisms:

$$\begin{aligned} \Phi_v &: v^* \mathcal{H}om(X, Y) \longrightarrow \mathcal{H}om(v^* X, v^* Y), & X, Y \in \mathrm{ob} E', \\ \Psi_v &: v_* \mathcal{H}om(X, Y) \longrightarrow \mathcal{H}om(v_* X, v_* Y), & X, Y \in \mathrm{ob} E, \\ \Xi_v &: \mathcal{H}om(v_! X, Y) \longrightarrow v_* \mathcal{H}om(X, v^* Y), & X \in \mathrm{ob} E, Y \in \mathrm{ob} E', \\ \Lambda_v &: v_!(X \times v^* Y) \longrightarrow v_!(X) \times Y, & X \in \mathrm{ob} E, Y \in \mathrm{ob} E'. \end{aligned}$$

In the definitions of Ξ_v and Λ_v one assumes that the inverse-image functor v^* has a left adjoint $v_!$.

10.5.1. Definition of Ψ_v

Let X be an object of E . The adjunction morphism $v^*v_*X \rightarrow X$ gives, for every object Z of E' , a bifunctorial morphism

$$v^*(Z \times v_*X) \longrightarrow v^*Z \times X.$$

Thus, for every object Y of E , one has a trifunctorial morphism

$$\mathrm{Hom}_E(v^*Z \times X, Y) \longrightarrow \mathrm{Hom}_E(v^*(Z \times v_*X), Y).$$

By adjunction this gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, v_*\mathcal{H}om(X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, \mathcal{H}om(v_*X, v_*Y)),$$

and hence the morphism Ψ_v .

10.5.2. Definition of Λ_v

For every object X of E there is an adjunction morphism $X \rightarrow v^*v_!X$. Therefore, for every object Y of E' , one obtains a bifunctorial morphism

$$X \times v^*Y \longrightarrow v^*v_!(X) \times v^*Y \simeq v^*(v_!(X) \times Y),$$

and hence, by adjunction, the morphism Λ_v .

10.5.3. Definition of Ξ_v

Let X be an object of E , and let Z, Y be objects of E' . The morphism

$$\Lambda_v : v_!(X \times v^*Z) \rightarrow v_!(X) \times Z$$

gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(v_!(X) \times Z, Y) \longrightarrow \mathrm{Hom}_{E'}(v_!(X \times v^*Z), Y).$$

By adjunction this gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, \mathcal{H}om(v_!X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, v_*\mathcal{H}om(X, v^*Y)),$$

and hence the morphism Ξ_v .

10.5.4. Definition of Φ_v

The trifunctorial morphism of 10.5.3,

$$\mathrm{Hom}_{E'}(v_!Z \times X, Y) \longrightarrow \mathrm{Hom}_{E'}(v_!(Z \times v^*X), Y),$$

where Z is an object of E and X, Y are objects of E' , gives by adjunction a trifunctorial morphism

$$\mathrm{Hom}_E(Z, v^*\mathcal{H}om(X, Y)) \longrightarrow \mathrm{Hom}_E(Z, \mathcal{H}om(v^*X, v^*Y)),$$

and hence one definition of Φ_v .

There is a second way to define Φ_v . Let X, Y, Z be three objects of E' . The functor $v^* : E' \rightarrow E$ gives a trifunctional morphism

$$\mathrm{Hom}_{E'}(Z \times X, Y) \longrightarrow \mathrm{Hom}_E(v^*Z \times v^*X, v^*Y).$$

By adjunction this yields

$$\mathrm{Hom}_{E'}(Z, \mathcal{H}om(X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, v_*\mathcal{H}om(v^*X, v^*Y)).$$

Thus one obtains a bifunctorial morphism

$$\mathcal{H}om(X, Y) \longrightarrow v_*\mathcal{H}om(v^*X, v^*Y),$$

and, again by adjunction, a morphism which the reader may verify is the same as the morphism Φ_v defined above.

Proposition 10.6. Let $v : E \rightarrow E'$ be a geometric morphism.

- 1) The morphism Ψ_v of 10.5.1 is an isomorphism for every object Y of E if and only if the adjunction morphism $v^*v_*X \rightarrow X$ is an isomorphism. In particular, Ψ_v is an isomorphism for every pair (X, Y) if and only if $v_* : E \rightarrow E'$ is fully faithful, i.e. if and only if v is an embedding of topoi.
- 2) The following conditions are equivalent:
 - (i) for every pair (X, Y) of objects of E' , the morphism Φ_v of 10.5.4 is an isomorphism;
 - (ii) the functor v^* has a left adjoint $v_!$ and, for every object X of E and every object Y of E' , the morphism Ξ_v of 10.5.3 is an isomorphism;
 - (iii) the functor v^* has a left adjoint $v_!$ and, for every object X of E and every object Y of E' , the morphism Λ_v of 10.5.2 is an isomorphism.

10.6.1

The first assertion follows immediately from 10.5.1. It also follows immediately from 10.5.2, 10.5.3, and 10.5.4 that (iii) is equivalent to (ii), and that these are equivalent to (i) together with the existence of a left adjoint $v_!$ to v^* . It remains to prove that (i) implies that v^* has a left adjoint, i.e. by IV, 1.8, that (i) implies that v^* commutes with small products. Let e' be the final object of E' and let I be a small set. Since v^* is left exact, $v^*(e')$ is final in E ; since v^* commutes with inductive limits, $v^*(\coprod_I e') \simeq \coprod_I v^*(e')$. For every object Y of E' one has

$$\mathcal{H}om(\coprod_I e', Y) \simeq \prod_I \mathcal{H}om(e', Y) \simeq \prod_I Y$$

and

$$\mathcal{H}om(\coprod_I v^*(e'), v^*Y) \simeq \prod_I \mathcal{H}om(v^*(e'), v^*Y) \simeq \prod_I v^*Y.$$

The functorial morphism Φ_v therefore induces an isomorphism

$$v^* \prod_I Y \simeq \prod_I v^*Y,$$

and one checks that this is the canonical isomorphism.

Corollary 10.7. Let E be a topos, let Z be an object of E , let $j_Z : E_{/Z} \rightarrow E$ be the localization morphism, and let X, Y be two objects of E .

1) There is a bifunctorial isomorphism

$$j_Z^* \mathcal{H}om(X, Y) \simeq \mathcal{H}om(j_Z^* X, j_Z^* Y).$$

2) If X' is an object of $E_{/Z}$, there is a bifunctorial isomorphism

$$\mathcal{H}om_E(j_{Z!} X', Y) \simeq j_{Z*} \mathcal{H}om_{E_{/Z}}(X', j_Z^* Y).$$

3) If Y' is an object of $E_{/Z}$ and Z is an open of E , then there is a bifunctorial isomorphism

$$j_{Z*} \mathcal{H}om(X', Y') \simeq \mathcal{H}om(j_{Z*} X', j_{Z*} Y').$$

Assertions 1) and 2) are proved by observing that Λ_{j_Z} is an isomorphism. For 3) it is enough to observe that j_{Z*} is fully faithful, because $j_{Z!}$ is fully faithful (I, 5.7 a)).

Corollary 10.8. Let E be a topos, let X, Y, Z be three objects of E , and let $j_Z : E_{/Z} \rightarrow E$ be the localization morphism.

1) There is a canonical isomorphism

$$j_{Z*} j_Z^* Y \simeq \mathcal{H}om(Z, Y).$$

2) There is a canonical isomorphism

$$\mathrm{Hom}_E(Z, \mathcal{H}om(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y).$$

Let e_Z be the final object of $E_{/Z}$, i.e. the object $\mathrm{id}_Z : Z \rightarrow Z$. From (10.2.1) one has a canonical isomorphism

$$\mathcal{H}om_{E_{/Z}}(e_Z, j_Z^* Y) \simeq j_Z^* Y.$$

By Corollary 10.7, 2), this gives an isomorphism

$$j_{Z*} j_Z^* Y \simeq \mathcal{H}om(j_{Z!} e_Z, Y).$$

Since $j_{Z!} e_Z = Z$, this proves 1). To prove 2), Corollary 10.7 gives a canonical isomorphism

$$\mathrm{Hom}_{E_{/Z}}(e_Z, j_Z^* \mathcal{H}om(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y).$$

Adjunction on the first member then gives the asserted isomorphism.